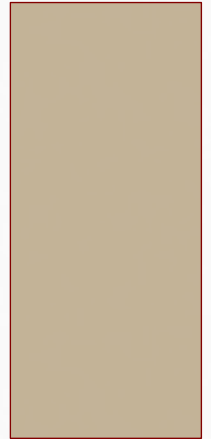


LIGHTING

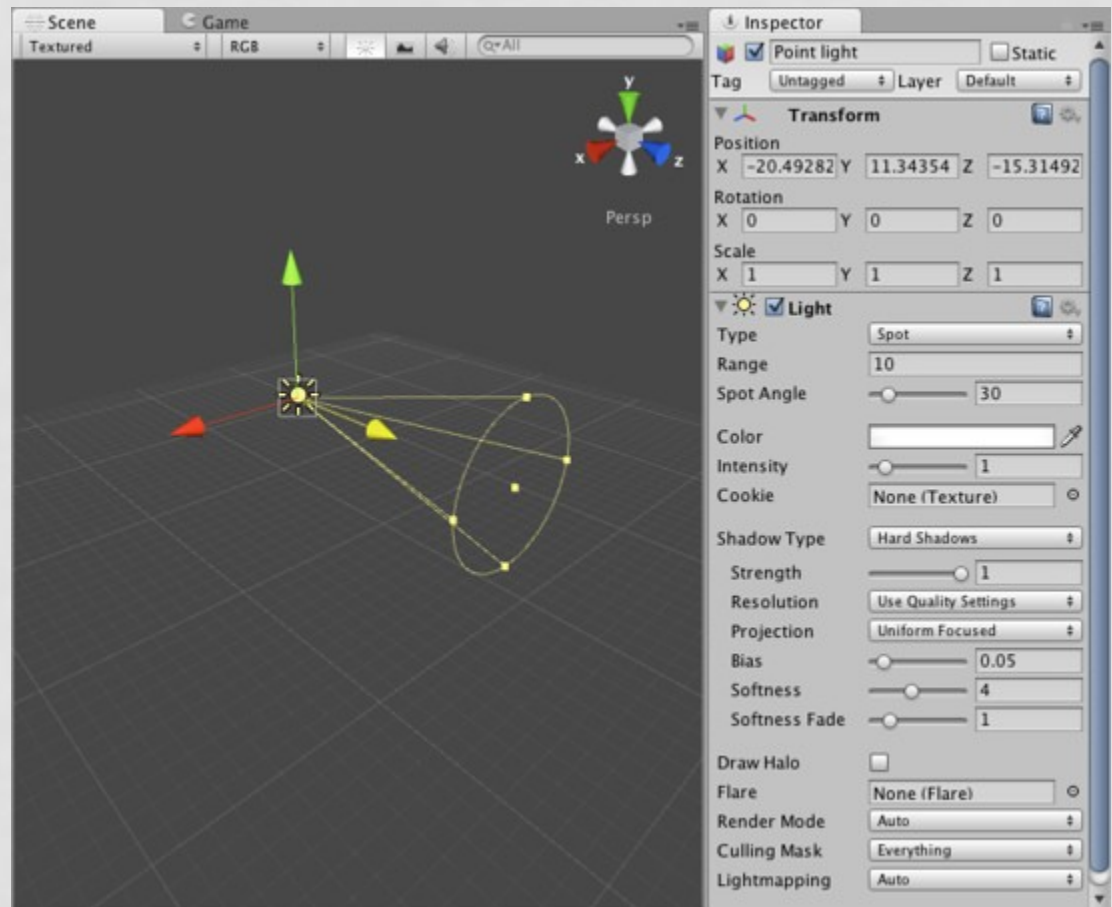
ADVANCED GAME DESIGN & DEVELOPMENT

DANNY JUGAN



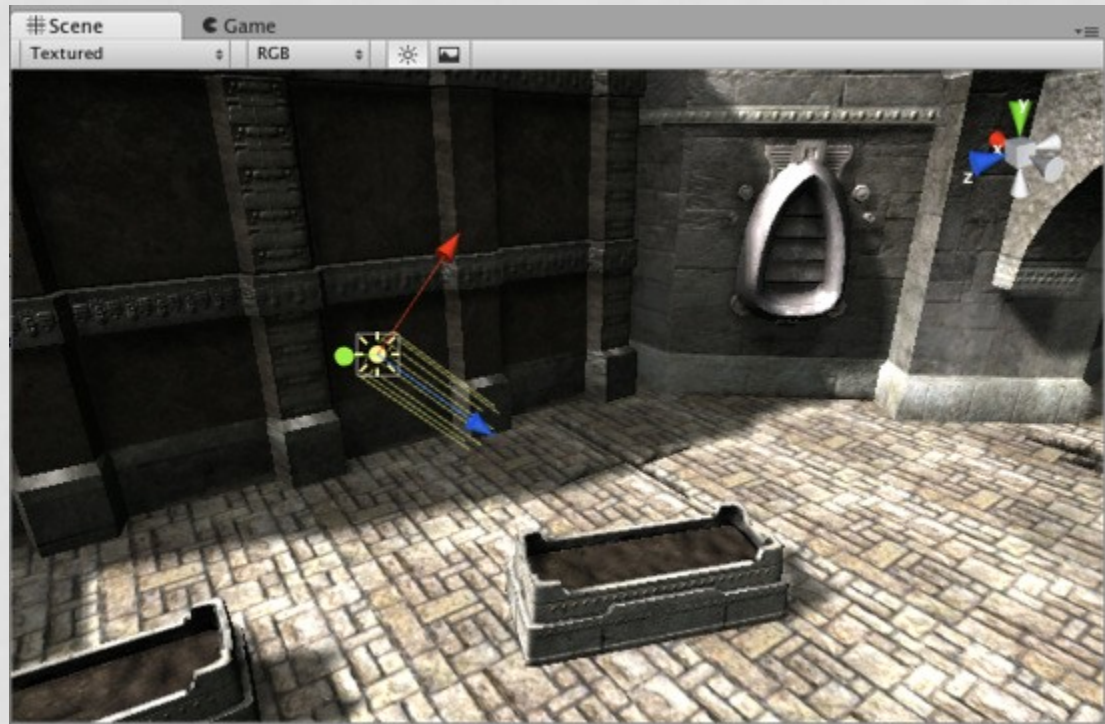
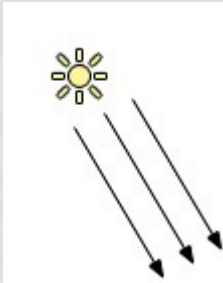
LIGHTING

- Lights will bring personality and flavor to your game.
- Use lights to illuminate the scenes and create the perfect visual mood.



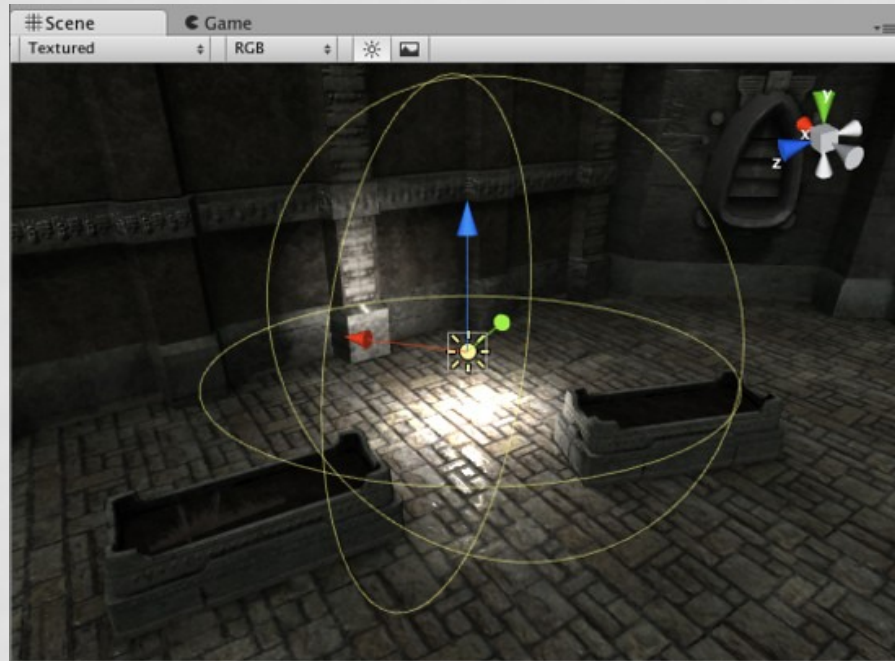
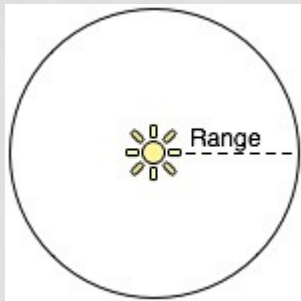
DIRECTIONAL LIGHTS

- **Directional lights** are placed infinitely far away and affect everything in the scene, like the sun.
 - Used mainly in outdoor scenes for sun & moonlight



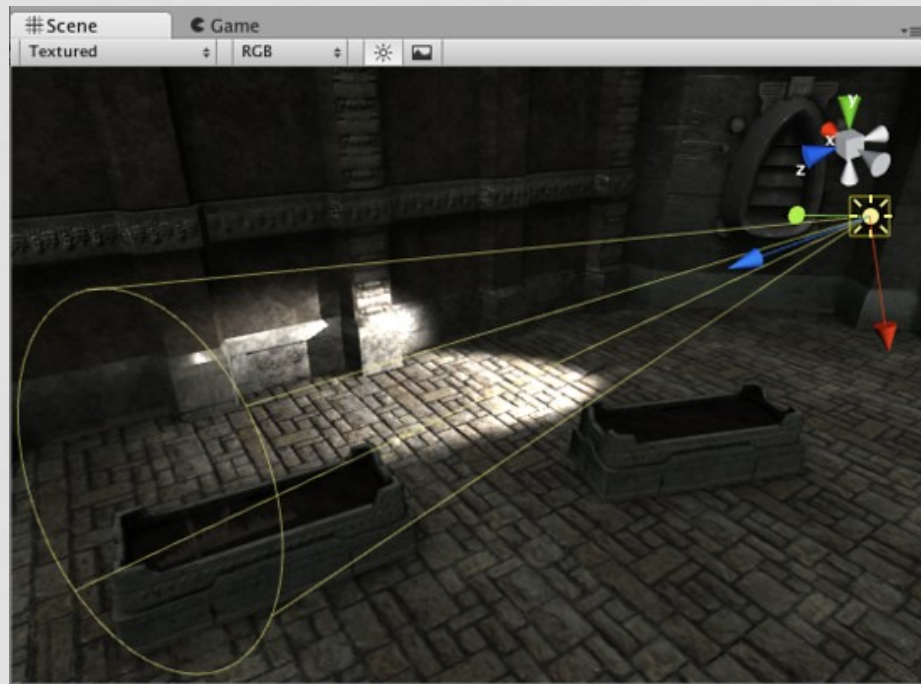
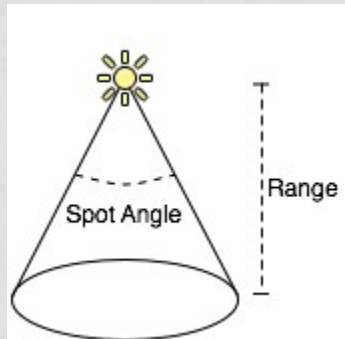
POINT LIGHTS

- **Point lights** shine from a location equally in all directions, like a light bulb.
 - They are the most common lights in computer games - typically used for explosions, light bulbs, etc.



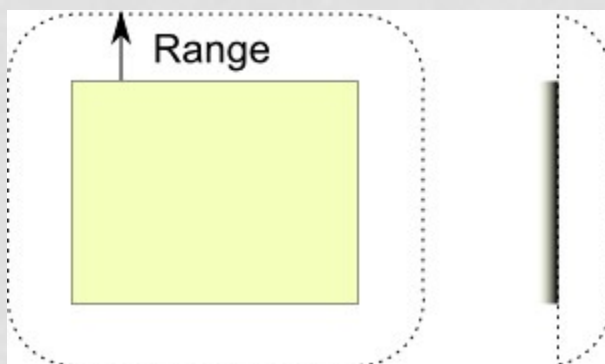
SPOT LIGHTS

- **Spot lights** shine from a point in a direction and only illuminate objects within a cone - like the headlights of a car.
 - They are perfect for flashlights, car headlights or lamp posts



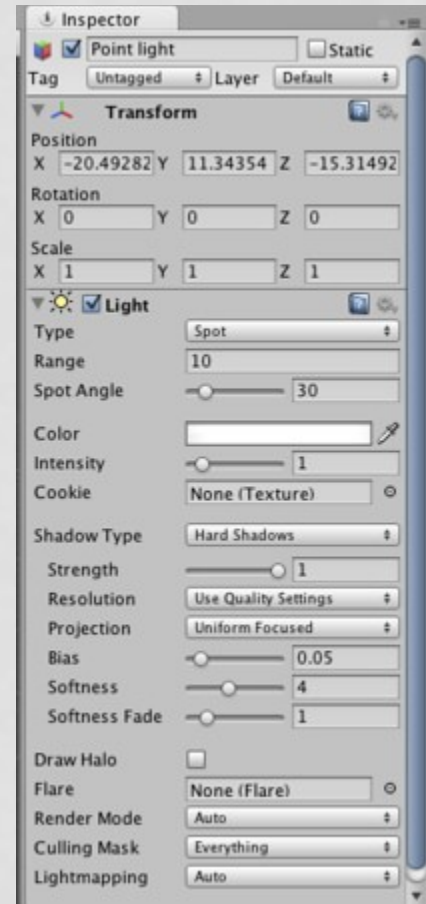
AREA LIGHTS

- **Area lights** (only available for lightmap baking) shine in all directions to one side of a rectangular section of a plane.
 - Area lights cast light from one side of a rectangular area of a plane.



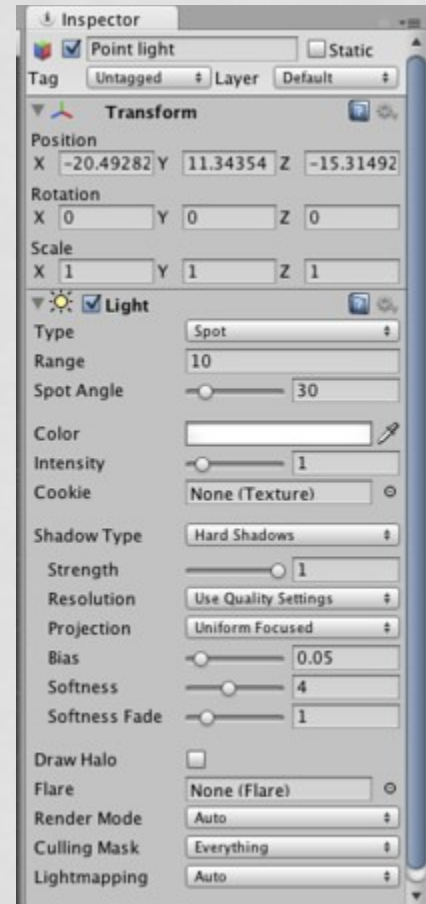
LIGHT PROPERTIES

- **Type:** The current type of light object:
 - **Directional:** A light placed infinitely far away. It affects everything in the scene.
 - **Point:** A light that shines equally in all directions from its location, affecting all objects within its Range.
 - **Spot:** A light that shines everywhere within a cone defined by Spot Angle and Range. Only objects within this region are affected by the light.
 - **Area:** A light that shines in all directions to one side of a rectangular area of a plane. The rectangle is defined by the Width and Height properties. Area lights are only available during lightmap baking and have no effect on objects at runtime.
- **Range:** How far light is emitted from the center of the object. Point/Spot light only.
- **Spot Angle:** Determines the angle of the cone in degrees. Spot light only.
- **Color:** The color of the light emitted.



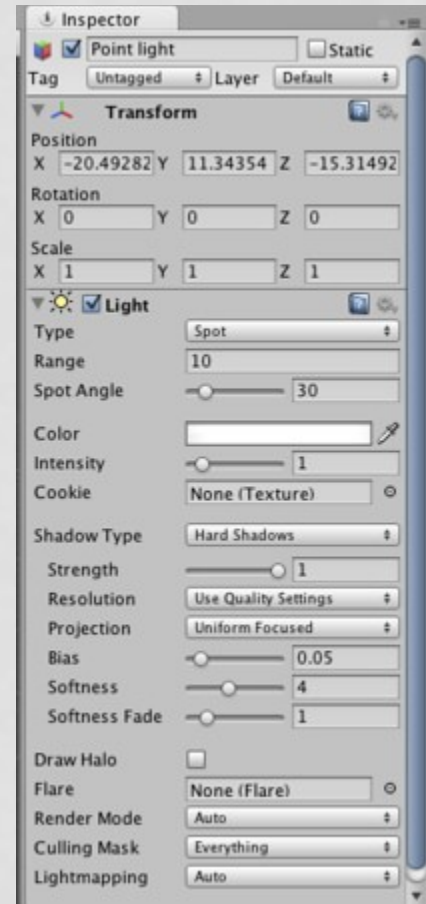
LIGHT PROPERTIES

- **Intensity:** Brightness of the light. Default value for a Point/Spot/Area light is 1. Default value for a Directional light is 0.5.
- **Cookie:** The alpha channel of this texture is used as a mask that determines how bright the light is at different places. If the light is a Spot or a Directional light, this must be a 2D texture. If the light is a Point light, it must be a Cubemap.
- **Cookie Size:** Scales the projection of a Cookie.
- **Draw Halo:** If checked, a spherical halo of light will be drawn with a radius equal to Range. See also the [Halo](#) documentation and the [Render Settings](#) documentation for more details.
- **Flare:** Optional reference to the [Flare](#) that will be rendered at the light's position.



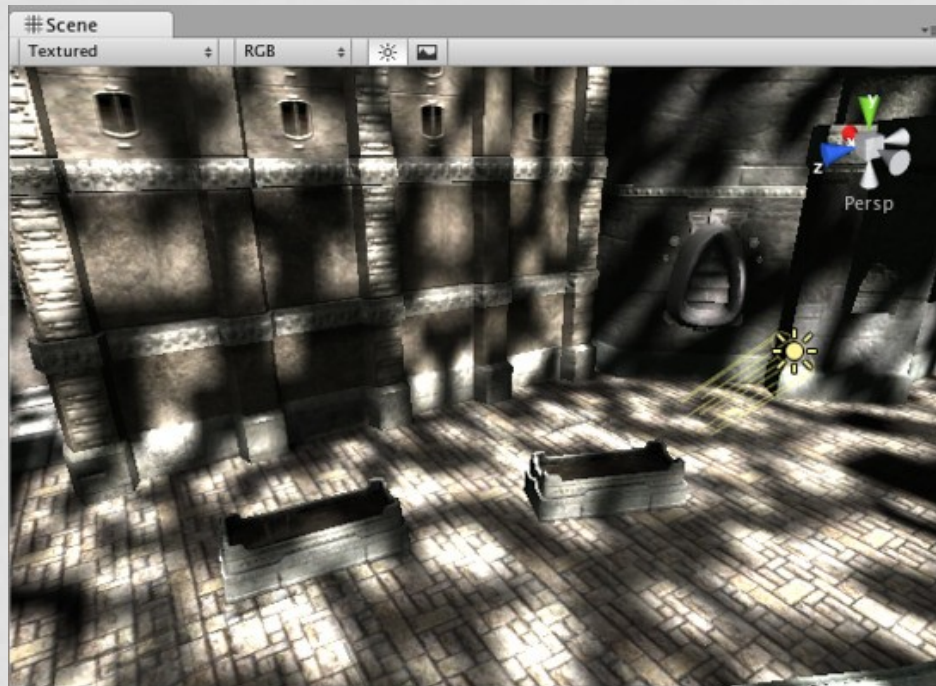
LIGHT PROPERTIES

- **Render Mode:** Importance of this light. This can affect lighting fidelity and performance, see *Performance Considerations* below. Options include:
 - **Auto:** The rendering method is determined at runtime depending on the brightness of nearby lights and current [Quality Settings](#).
 - **Important:** This light is always rendered at per-pixel quality. Use this for very important effects only (e.g. headlights of a player's car).
 - **Not Important:** This light is always rendered in a faster, vertex/object light mode.
- **Culling Mask:** Use to selectively exclude groups of objects from being affected by the light; see [Layers](#).
- **Lightmapping:** The Lightmapping mode: Realtime Only, Auto or Baked Only; see the [Dual Lightmaps](#) description.
- **Width:** (Area lights only) The width of the rectangular light area.
- **Height:** (Area lights only) The height of the rectangular light area.



COOKIES

- Directional Light projection a cloud-like cookie texture.
 - A cookie is a great way to add some quick detail to large outdoor scenes.
 - You can even slide the light slowly over the scene to give the impression of moving clouds.



LIGHTING HELP

- Great Tutorials
 - <http://www.youtube.com/watch?v=u5DNkxkXBeI>
 - <http://www.youtube.com/watch?v=TZPdz-M476M>
 - <http://unity3d.com/learn/tutorials/modules/beginner/live-training-archive/lights>
 - <http://unity3d.com/learn/tutorials/projects/stealth/game-setup-lighting>